SYSDESIGN REFERENCE

Message Queues Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Decouple producers from consumers and absorb bursts with explicit delivery semantics.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / MESSAGE-QUEUES

SYSDESIGN

Message Queues Tradeoff Matrix

CORE IDEA

Decouple producers from consumers and absorb bursts with explicit delivery semantics.

Reference architecture decision chain

01	02	03	04
Ordering	Partitions	Retention	Retries

Decisions to defend

+ Ordering
+ partitions
+ retention
+ retries
+ dead letters
+ consumer groups
+ and idempotency.

Failure modes to test

+ Poison messages
+ backlog growth
+ duplicate effects
+ and reordering.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Message Queues Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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